

LEARN ONE THING
EVERY DAY

MAY 1 1918

SERIAL NO. 154

THE MENTOR

THE STORY OF
COAL

By
CHARLES FITZHUGH TALMAN
Editorial Writer for the
Scientific American

DEPARTMENT OF
SCIENCE

VOLUME 6
NUMBER 6

TWENTY CENTS A COPY

LEARN ONE THING
EVERY DAY

MAY 1 1918

SERIAL NO. 154

THE MENTOR

THE STORY OF
COAL

By
CHARLES FITZHUGH TALMAN
Editorial Writer for the
Scientific American

DEPARTMENT OF
SCIENCE

VOLUME 6
NUMBER 6

TWENTY CENTS A COPY

**The Project Gutenberg eBook of The Mentor: The Story of Coal, vol. 6,
Num. 6, Serial No. 154, May 1, 1918**

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The Mentor: The Story of Coal, vol. 6, Num. 6, Serial No. 154, May 1, 1918

Author: Charles Fitzhugh Talman

Release date: February 2, 2016 [eBook #51106]

Most recently updated: October 22, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/51106

Credits: Produced by Juliet Sutherland and the Online Distributed
Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK THE MENTOR: THE STORY
OF COAL, VOL. 6, NUM. 6, SERIAL NO. 154, MAY 1, 1918 ***

**THE MENTOR 1918.05.01, No. 154,
The Story of Coal**

LEARN ONE THING
EVERY DAY

MAY 1 1918

SERIAL NO. 154

THE
MENTOR

THE STORY OF COAL

By
CHARLES FITZHUGH TALMAN

Editorial Writer for the
Scientific American

DEPARTMENT OF
SCIENCE

VOLUME 6
NUMBER 6

TWENTY CENTS A COPY

THE MINER

By BERTON BRALEY

Grimy, and caked with dust of coal he stands,
Grasping his pick within his mighty hands;
The arbiter of destiny and fate,
Greater by far than king or potentate.

Shops may not run except at his behest,
At forge and blast his strength is manifest.
The rolls that rumble and the shears that scream
And all the million miracles of steam

Depend on him for fuel that will turn
The wheels that urge them and the belts that churn.
Guns that will shatter fortresses of steel,
Ships that will plow the waves on steady keel

Bearing munitions for an army's need
Must wait the miner's orders and take heed
That he who toils within the coal mine's murk
Gives them the coal with which they do their work.

Behind the men who battle in the trench
There stand the workmen at the lathe and bench,
But back of them and master of them all
The miner stands and holds the world in thrall.

Not soon again shall any man forget
How much the world is in the miner's debt,
For we shall read upon fame's honor roll
"He won the war—his labor gave us coal!"

Reprinted by courtesy of Publishers of "Coal Age."



COURTESY U. S. GEOLOGICAL SURVEY

FOSSIL FERN FROM COAL MINE

THE STORY OF COAL

The Origin of Coal

ONE



While the vegetable origin of coal is beyond question, two rival views are current among geologists to account for the deposit of ancient plant material in the form of coal-beds, such as we now find in the earth. One school of geologists holds that the coal plants grew in great lagoons and swamps, like the mangrove swamps of today, and that the modern coal-beds mark the locations of these swamps. From time to time these areas subsided and were flooded with water to such a depth that the plants were killed. Eventually the decayed vegetation of the former swamps was covered with a layer of mud or sand. Later a slow upheaval of the ground brought these regions again to the surface; a new swamp formed, only to be submerged again at a later period; and the same process was repeated several times in the course of hundreds of thousands of years.

The bulk of evidence seems to favor this view, but there is another. Perhaps the coal-beds are not the sites of former swamps, but of estuaries and ocean shores where the plant material settled down, in still water, after a long drift down the ancient rivers from its place of origin. It is not impossible that both explanations are correct; some coal-beds having been formed in one way, and some in the other. With the progress of time the deposits of sand were compacted into sandstone, and the mud and clay into shale; while the layers of vegetation were solidified by pressure, some of their constituents were vaporized and expelled by heat, and the final product was coal.

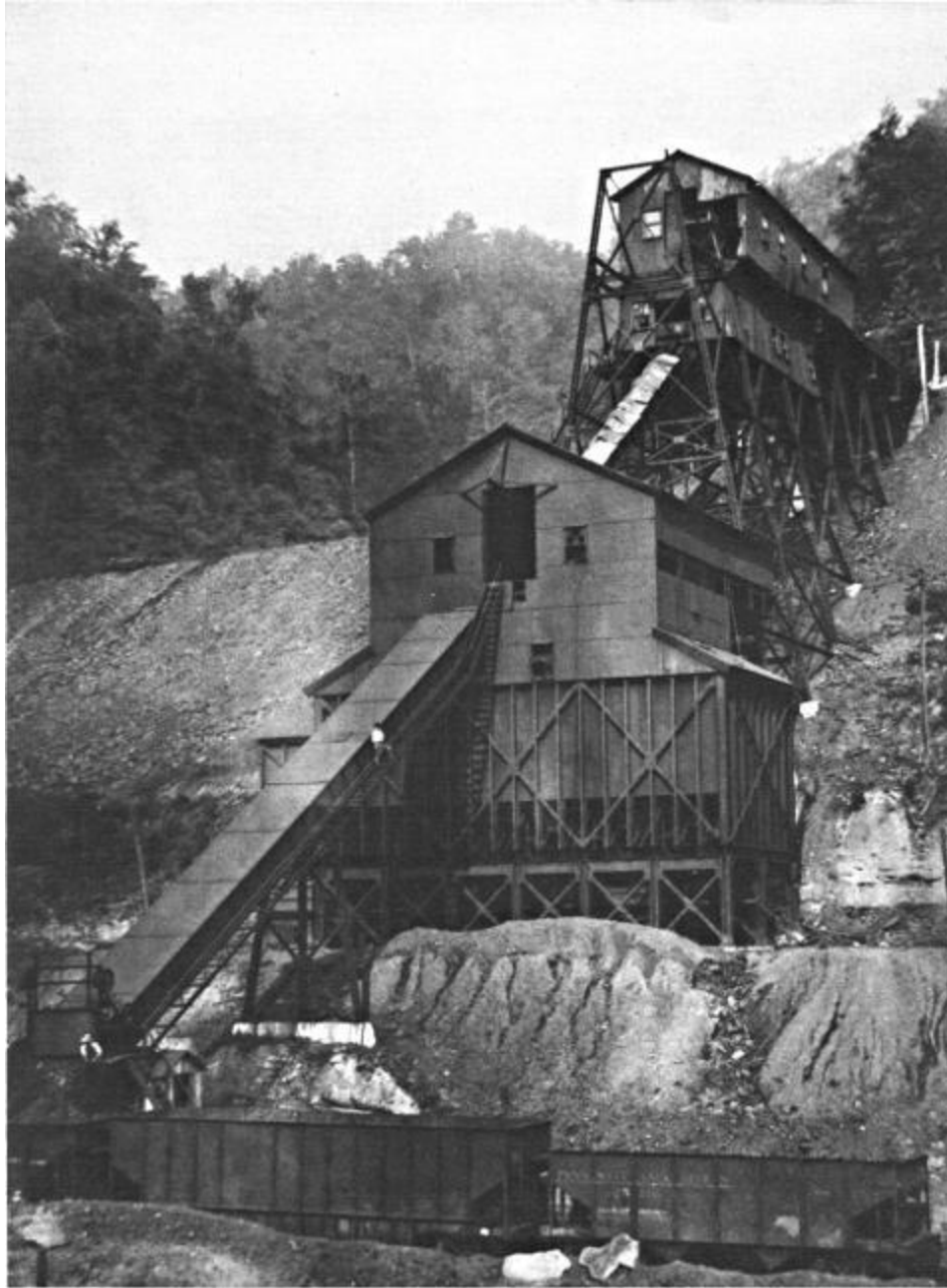
The coal-measures abound in fossil plants of species long ago extinct, and we also find the molds or casts of plants that have themselves disappeared, leaving only their impressions in the mud by which they were

once enveloped. These records of ancient vegetation are mostly found in the rocks just above and below the coal-beds, and not in the coal itself.

The plants of the Carboniferous Period, during which most but not all of the coal-beds were formed, bore a family likeness to certain kinds of plants that flourish today. Many of them were ferns, ranging in size from the smallest species up to great tree-ferns. Others resembled our modern horsetails or scouring-rushes, with their fluted and jointed stems, but these *calamites*, as the geologists call them, grew to the size of trees, sometimes eighty or ninety feet in height. Some plants of the coal age were like the modern cycads (intermediate in appearance between tree-ferns and palms); some were like the ginkgo, a tree with leaves like those of maidenhair fern, widely introduced into this country from China and Japan. One of the commonest and largest trees was the *lepidodendron*, closely resembling, except in its vastly greater size, the club-moss or ground pine which we know so well as a Christmas decoration.

The animal life of the period, of which, also, abundant fossil remains are found, included mollusks, fishes, crustaceans, insects, spiders, thousand-legs, snails, reptiles and lizards. Some of the insects were a foot or more in length. Of cockroaches, alone, more than five hundred species have been found in the coal-measures.

PREPARED BY THE EDITORIAL STAFF OF THE MENTOR ASSOCIATION
ILLUSTRATION FOR THE MENTOR. VOL. 6, No. 6, SERIAL No. 154
COPYRIGHT, 1918. BY THE MENTOR ASSOCIATION, INC.



COURTESY U. S. BUREAU OF MINES

TIPPLE AT BITUMINOUS COAL MINE GARY, WEST VIRGINIA

THE STORY OF COAL

The Coal Fields of the United States

TWO



When a coal famine is upon us there is a grain of comfort in the reflection that beneath the soil of this country, and within 3,000 feet of the surface, there still lies 3,538,554,000,000 tons of coal. This is the estimate of the United States Geological Survey. We have mined coal wastefully and used it prodigally, yet we have taken from the ground, up to the present time, only a fraction of one per cent. of the total amount at our disposal. The whole of our “coal reserves,” if they could be extracted and placed in a great cubical pile, would form a mass 8.4 miles long, 8.4 miles wide and 8.4 miles high. If the coal thus far mined were piled up in the same way, the cube would be 7,200 feet long, 7,200 feet wide, and 7,200 feet high. ^[1]

^[1] These figures were furnished by Mr. M. R. Campbell, of the U. S. Geological Survey. They differ materially from figures previously published by the Survey.

The coal-producing areas of the country are divided into six great divisions, known as the Eastern Province, the Interior Province, the Gulf Province, the Northern Great Plains Province, the Rocky Mountain Province, and the Pacific Coast Province. The Eastern Province contains probably nine-tenths of the high-rank coal of the country. It is made up of the anthracite regions of Pennsylvania and Rhode Island, the Atlantic coast region of Virginia and North Carolina, and the great Appalachian region, which embraces all the bituminous and semi-bituminous coal of what is called the “Appalachian trough.” The state of Pennsylvania produces 47 per cent. of all the coal mined in the country, and nearly all of the anthracite.

The Appalachian region is the greatest storehouse of high-rank coal in the United States, if not in the world. “This near-by and almost inexhaustible supply of high-grade fuel,” says the Geological Survey, “has been the foundation of the development of the blast furnaces, the great iron and steel mills, and the countless manufacturing enterprises of the Eastern states.”

The Interior Province includes all the bituminous coal fields and regions near the Great Lakes, in the Mississippi Valley, and in Texas, and is made up of four distinct sections—the northern (Michigan), eastern (Illinois, Indiana and western Kentucky), western (Iowa, Missouri, Kansas, Oklahoma and Arkansas), and southwestern (Texas). The coal of this province is not, in general, of as high a quality as that of the Eastern Province, but it is very extensively mined, and is used for heating and for generating power in the many cities and towns of the Mississippi valley and the Great Lakes region. Indeed, extensive coal fields in proximity to rich agricultural lands have made possible the existence of such manufacturing centers as Chicago, St. Louis and Kansas City, and have been a leading factor in the development of the vast railway systems of the Middle West.

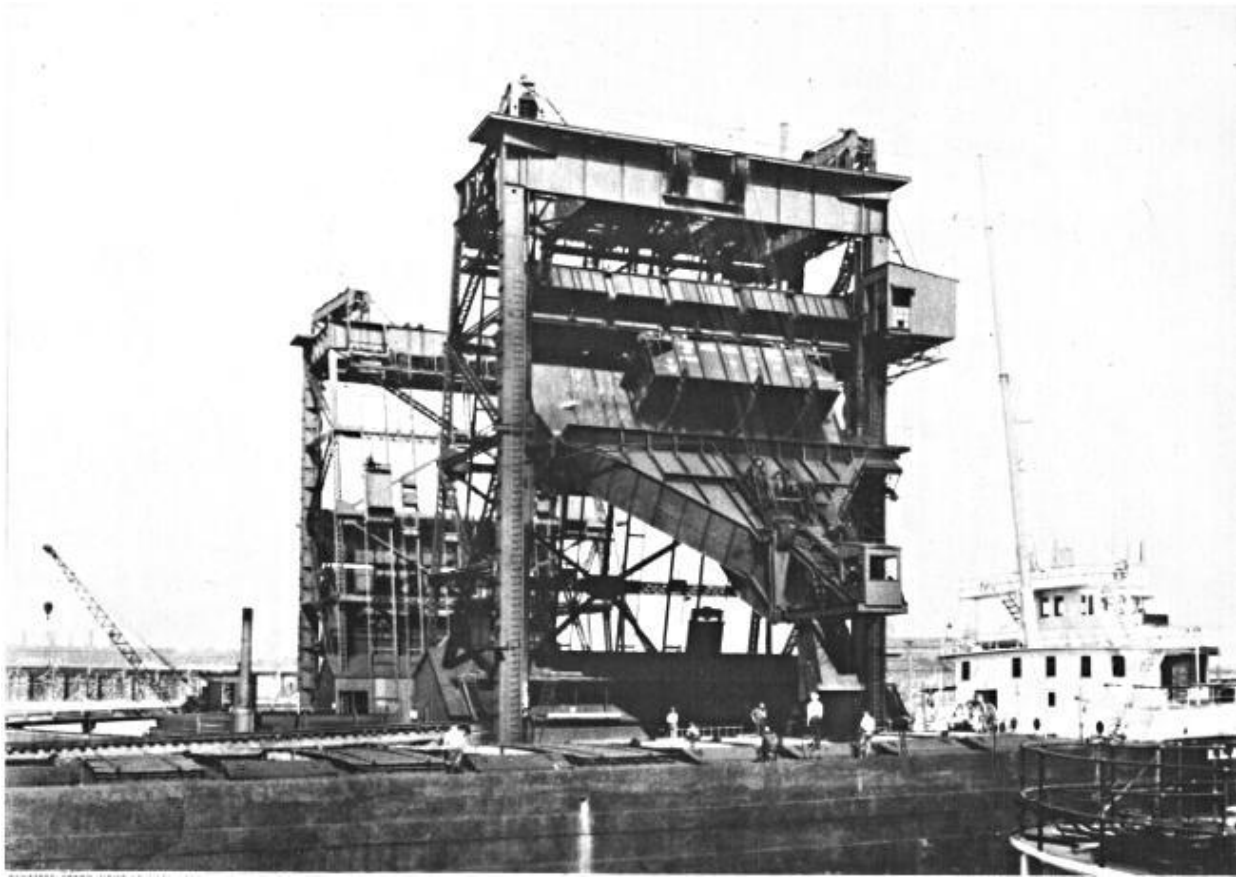
The Gulf Province is at present of little commercial importance. Its coal is mined only at a few places in Texas, and is mostly lignite.

The Northern Great Plains Province includes all the coal fields in the Great Plains east of the Front Range of the Rocky Mountains. The coals are of low rank, being either lignite or sub-bituminous, except in a few of the basins near the mountains. The largest coal region in this province is the Fort Union region, lying in the Dakotas, Montana and Wyoming. The amount of unmined coal in this region is estimated to be twice as great as that lying in the rich Appalachian region, but it has been little worked, as it is generally of poor quality.

The Rocky Mountain Province contains a greater variety of coal than any other province in the United States. It includes all ranks, from lignite to anthracite, but the prevailing ranks are sub-bituminous and low-grade bituminous.

The coal of the Pacific Coast Province is mined chiefly in the state of Washington, where it has aided in developing the industries of the Puget Sound region. Oregon and California have small fields, but the coal is of poor quality, and little mining has been attempted.

PREPARED BY THE EDITORIAL STAFF OF THE MENTOR ASSOCIATION
ILLUSTRATION FOR THE MENTOR. VOL. 6, No. 6, SERIAL No. 154
COPYRIGHT, 1918. BY THE MENTOR ASSOCIATION, INC.



COURTESY BROWN HOISTING MACHINERY CO. CLEVELAND, O.
COAL CAR DUMPER IN OPERATION

THE STORY OF COAL

Handling Coal

THREE



In times gone by coal was carried out of the mines on the shoulders of men and women, and then transferred in wheelbarrows to the sailing-vessels or wagons in which it was taken to market. In progressive mines of today the coal is loaded in the mine into small mine cars, which are hauled and hoisted to the surface by electricity or steam. The mine cars are dumped on an elevated platform called the "tipple," and the coal passes through chutes or conveyors to the railway cars waiting underneath to receive it. On its way downward it undergoes a more or less elaborate process of screening, breaking, picking, washing, etc., according to the kind of coal and the purpose for which it is to be used.

The coal reaches the market by three general methods of transportation: (1) All-rail; (2) rail to the seaports, where it is used for bunkering steamers or carried by vessels to other ports, foreign and domestic; (3) rail to the Great Lakes, especially Lake Erie ports, from which it is carried to ports on the upper lakes, and from the latter again by rail to markets in the interior. The railroads themselves use about one-fourth of all the coal mined in this country. The coastwise coal-carrying trade is mainly by wooden barges towed by steamers, though much coal is also carried by schooners, some of which can carry a cargo of 5,000 tons or more. About four per cent. of the bituminous coal output goes to foreign countries.

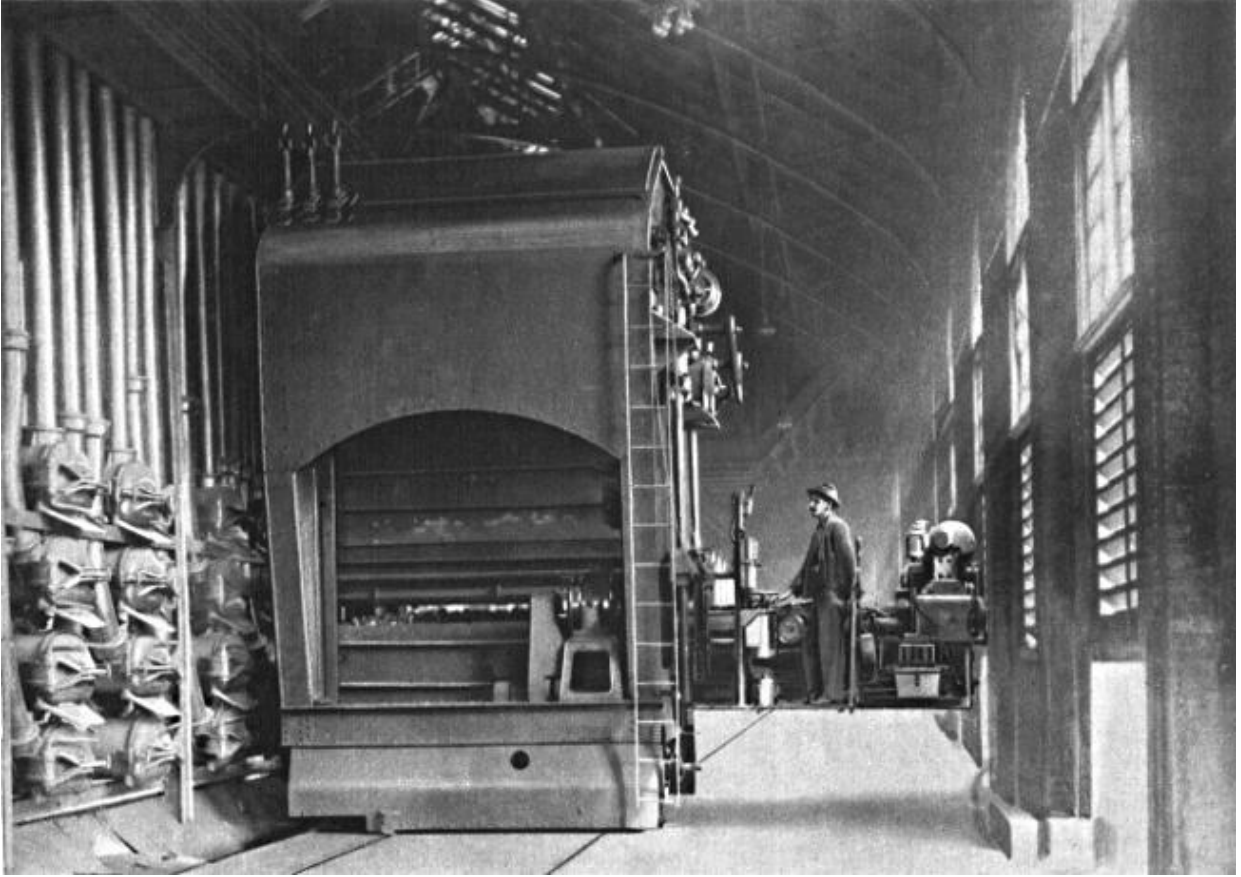
"The consumption of coal," says the United States Geological Survey, "is a measure of the industrial activity of a people, for as yet coal is the main source of mechanical energy. In this respect the United States is the foremost nation, its average annual consumption of coal for all purposes being about five tons per capita. Prior to the present war in Europe the consumption of coal per capita in England, Belgium and Germany was about four tons, in Russia a quarter of a ton, and in France about 1.6 tons."

Marvelous forms of labor-saving machinery have been introduced to facilitate the loading and unloading of coal. The principal form of apparatus for

transferring coal either to or from a vessel is the “bridge tramway plant,” which consists of long steel bridges mounted side by side on suitable rails so that they can be moved into place over the hatchway of a vessel. Huge buckets, which load and unload themselves, are carried on a “trolley,” suspended from the bridge, and transfer coal at high speed from the vessel to the stock pile or railway cars, or *vice versa*. The cost of loading coal by this method is only a cent or two a ton.

Another ingenious device is the “car-dumper.” This powerful machine picks up bodily from the railway track a car loaded with a hundred tons of coal, overturns it, and discharges its contents into the hold of a vessel; after which it returns the car to the track. It is capable of handling fifty cars an hour. It is equipped with special apparatus to prevent the coal from being discharged too violently, and thus being badly broken up.

PREPARED BY THE EDITORIAL STAFF OF THE MENTOR ASSOCIATION
ILLUSTRATION FOR THE MENTOR. VOL. 6, No. 6, SERIAL No. 154
COPYRIGHT, 1918. BY THE MENTOR ASSOCIATION, INC.



COURTESY POPULAR SCIENCE MONTHLY

CHARGING COAL IN A MODERN GAS PLANT

THE STORY OF COAL

Coal Products

FOUR



he story of the coal products forms one of the most romantic chapters in the history of applied science. The marvels of fairyland are surpassed by the achievements of the modern manufacturer in obtaining from mere black rocks dug out of the ground not only heat and light, but a bewildering variety of useful gases, liquids and solids—drugs, chemicals, dyestuffs, and so forth.

For hundreds of years it has been known that when coal is covered or enclosed, to keep out the air, and then heated for a certain length of time, instead of burning to ash it is converted into a porous grayish-black substance called “coke.” This material, which burns without smoke or flame, is a valuable fuel for many purposes; especially for use in blast-furnaces for the smelting of ore. Nowadays coke is made on a vast scale from certain grades of bituminous and semi-bituminous coal. The coal is heated in “coking-ovens,” of which there are several kinds. The most common form of oven in this country is the “bee-hive oven,” which produces coke only. Another type of coking-oven, more generally used in Europe than in America, is the “flue-oven,” which produces, besides coke, a number of valuable by-products.

When coal is converted into coke it gives off combustible gases. The idea of saving these gases and using them for illuminating purposes was first practically applied in the latter part of the eighteenth century. “Coal-gas” is made by heating coal in a closed vessel, called a “retort.” It is a mixture of hydrogen and methane (a compound of hydrogen and carbon), with small amounts of several other gases. Most of the carbon in the coal remains in the retort as coke, which is, therefore, a by-product in the process of making coal-gas. After the gas is given off from the coal it passes through a series of vessels, where, by chemical and other methods, it is freed from ingredients which would impair its value as an illuminant, but which are saved and used for other purposes; the most important of these are “coal-tar” and “ammoniacal liquor.”

The purified coal-gas is finally conveyed to a gas-holder or “gasometer,” from which it is distributed to the consumers.

In recent times other methods of gas-making have come into use. In one of these nearly all the carbon in the fuel is turned into a combustible gas by passing air through the hot coal. The product is known as “producer-gas,” and is very valuable for use as fuel and as a motive power in gas-engines, but it is not an illuminant. A modification of this process, in which steam is passed over the heated fuel, gives a mixture of hydrogen and carbon monoxide, known as “water-gas.” This is also a valuable source of heat and power; but for use as an illuminant it must be mixed with a gas made from oil. It is then known as “carburetted water-gas,” and is very extensively used for lighting purposes; either by itself or mixed with ordinary coal-gas.

Of the by-products of gas-making, ammoniacal liquor was, until recently, the only commercial source of ammonia. Coal-tar, formerly thrown away as worthless, is today the source of innumerable substances of immense value to science and the industries. From coal-tar are obtained benzene, toluene, xylene, phenol (carbolic acid), naphthalene, anthracene, etc., and these more direct products are combined with one another or with other chemicals to produce coloring matters, explosives, perfumes, flavoring materials, sweetening substances, disinfectants, medicines, photographic developers—in short, a little of everything. The total number of coal-tar products runs into the thousands, and is constantly being increased by fresh discoveries.

In Germany, just before the war, the industries engaged in making these products (no longer *by-products*, but far more important than coke and gas) were capitalized at \$750,000,000. One firm made no less than 1,800 coal-tar dyes, besides 120 pharmaceutical and photographic preparations.

PREPARED BY THE EDITORIAL STAFF OF THE MENTOR ASSOCIATION
ILLUSTRATION FOR THE MENTOR. VOL. 6, No. 6, SERIAL No. 154
COPYRIGHT, 1918. BY THE MENTOR ASSOCIATION, INC.



SMOKE PROBLEM—SCENE IN PITTSBURGH BEFORE AND AFTER THE SMOKE CURE

THE STORY OF COAL

The Smoke Problem

FIVE



Smoking is a costly and injurious habit. This is not the beginning of an appeal on behalf of the anti-cigarette crusade, but the introduction to a few facts in regard to the far-reaching effects of smoky chimneys. The smoke nuisance is as old as the use of coal.

In the fourteenth century a man was executed in London for befouling the air of that city with the fumes of “sea-coal,” as ordinary mineral coal was once called in England, because it was brought to London by sea. Under Queen Elizabeth a law was passed forbidding the burning of coal while Parliament was in session, as the legislators believed their health was likely to be impaired by the smoky air of the city. What would these bygone gentlemen say if they could see modern London enveloped in one of its famous “pea-soup” fogs—the color and denseness of which are entirely due to coal-smoke?

Smoke is injurious to health, destructive to vegetation, and fatal to architectural beauty; and, along with all this, it is enormously expensive. In the first place, a smoky chimney means imperfect combustion, and a waste of part of the heating value of fuel. Then a smoky atmosphere entails big laundry and dry-cleaning bills; frequent repainting of houses; injury to metal work; damage to goods in shops; excessive artificial lighting in the daytime. Pittsburgh was once the most famous American example of all these evils, but it has recently reformed. Before the Mellon Institute of Industrial Research carried out its elaborate smoke investigation in that city, and, in consequence, stringent smoke-abatement ordinances were adopted, the annual smoke bill of Pittsburgh was estimated at nearly ten million dollars. The city was the paradise of the laundryman, and light-colored clothing was so little worn by the inhabitants that it was known as “the mourning town.”

Throughout the United States it is said that smoke causes an annual waste and damage amounting to half a billion dollars. No wonder numerous societies have been formed to mitigate this evil, and a great many laws have been enacted on the subject. With a gradual increase in the use of gas, coke and other

smokeless fuels, and improved methods of stoking furnaces, the smoke nuisance is now happily abating.

The pollution of the air by smoke is the subject of systematic investigation and measurement at certain places in this country and abroad. Measurements of the “soot-fall” made in Pittsburgh a few years ago indicated an annual average deposit of soot in that city amounting to 1,031 tons per square mile. London’s average is 248 tons per square mile for the whole city and 426 tons in the central districts. In the heart of Glasgow the annual soot-fall is 820 tons per square mile.

In Great Britain there is a Committee for the Investigation of Atmospheric Pollution, which has installed standard measuring apparatus in sixteen English and Scotch towns. The soot is collected in a “pollution gauge,” consisting of a large cast-iron funnel, enameled on the inside. Projecting above the gauge is a wire screen, open at the top, to prevent birds from settling on the edge of the vessel. The gauge communicates at the bottom with one or more bottles for collecting rain-water, with its solid contents. The bottles are emptied once a month, when their contents are weighed and analyzed.

Smoke is injurious to the respiratory organs, conducive to eye-strain and responsible for a lowering of human vitality. The gloominess of a smoke-laden atmosphere also has a depressing effect upon the minds of many people.

PREPARED BY THE EDITORIAL STAFF OF THE MENTOR ASSOCIATION
ILLUSTRATION FOR THE MENTOR. VOL. 6, No. 6, SERIAL No. 154
COPYRIGHT, 1918. BY THE MENTOR ASSOCIATION, INC.



COURTESY U. S. BUREAU OF MINES

RESCUE PARTY ENTERING MINE AFTER EXPLOSION

THE STORY OF COAL

Safety in Coal Mines

SIX



ince the year 1870, some 60,000 men have lost their lives as a result of coal mining accidents in this country. This is approximately one fatality for every 180,000 tons of coal mined. Gradually this bad record is being improved, thanks to the combined efforts of the United States Bureau of Mines, the mining departments of the various states, the operators and the miners themselves; but coal mining remains a hazardous pursuit.

Falls of the roof are responsible for more accidents than any other single cause. These are likely to occur wherever the roof is not fully timbered; especially in the “rooms,” where the coal is being blasted out. Many accidents also occur in mine shafts, notwithstanding the various safety devices with which the “cage” or elevator is nowadays provided.

Fires and explosions attract a greater amount of public attention than other mining disasters on account of the large number of victims so often involved in a single occurrence of this kind. In the explosion at the Courrières colliery, in France, March 10, 1906, more than 1,100 miners perished. This mine had previously been renowned for its freedom from accidents. Coal mine explosions are due to two principal causes, which may act either separately or in combination—fire-damp and coal-dust. Accumulations of fire-damp, or methane, locked up in the coal seams, are liberated by the removal of the coal. Frequently streams of this gas gush forth with a hissing noise, and are known as “blowers.” Fire-damp is explosive when combined with certain proportions of air. Apart from ventilation, which dilutes the gas below the danger limit, the principal precaution against explosives is the use of safety-lamps, so constructed that the gas cannot come in contact with a naked flame. An excessive amount of coal-dust in the air of the mine may also give rise to explosions. Such explosions may be prevented by wetting the dust, moistening the air, or powdering the walls, roof and floor with a non-explosive “rock-dust.”

After an explosion the air of a mine contains a large amount of the deadly gas carbon monoxide, and this “after-damp,” as it is called, makes rescue work extremely dangerous. Wherever suitable apparatus is available, the rescuers carry with them a supply of oxygen, by breathing which they are able to live for some hours in a poisonous atmosphere. The Bureau of Mines has established a number of rescue stations in the coal-mining districts and maintains several mine safety cars for hurrying rescue crews to the scene of a disaster. The Bureau also instructs the miners in first-aid and rescue work, and is directing a national campaign in behalf of “safety first” in mines.

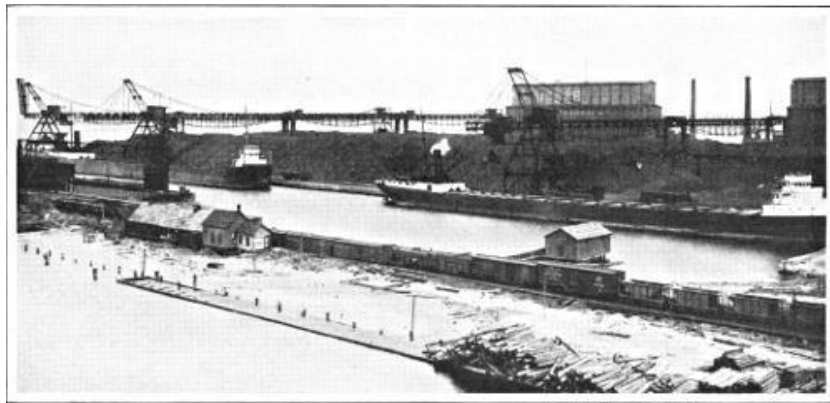
Of the many methods that have been devised for testing the air of mines for noxious gases none is more interesting than the use of caged canaries. These birds are much more

susceptible than human beings to the effects of carbon monoxide, and show signs of distress before a man begins to feel any discomfort from the gas. In many mines they are carried in routine inspections. After an explosion the number of rescuers equipped with oxygen apparatus is always limited. These form the advance guard, and are followed by men without apparatus, who carry canaries, by observing the behavior of which they can tell how far they may safely penetrate into the mine. The Bureau of Mines has devised a special form of cage in which the canary may be revived with oxygen after being overcome with gas. Experiments show that the bird may be asphyxiated and revived again and again without suffering any ill-effects; neither does he acquire an immunity to poisoning which would make him a less reliable indicator.

PREPARED BY THE EDITORIAL STAFF OF THE MENTOR ASSOCIATION
ILLUSTRATION FOR THE MENTOR. VOL. 6, No. 6, SERIAL No. 154
COPYRIGHT, 1918. BY THE MENTOR ASSOCIATION, INC.

THE MENTOR · DEPARTMENT OF SCIENCE

MAY 1, 1918



Courtesy of Brown Hoisting Machinery Co.

COAL-HANDLING MACHINERY ON PITTSBURGH COAL CO. DOCKS,
DULUTH, MINN.

An electric trolley, with 5½-ton bucket, travels on each bridge, carrying coal from the boat to storage, or to railroad car, or to screening apparatus in the rear

THE STORY OF COAL

By CHARLES FITZHUGH TALMAN

Editorial Writer for the Scientific American

MENTOR GRAVURES

FOSSIL FERN FROM COAL MINE · TIPPLE AT BITUMINOUS COAL MINE · COAL
CAR DUMPER IN OPERATION · CHARGING COAL IN A MODERN GAS PLANT ·
SMOKE PROBLEM, SCENE IN PITTSBURGH BEFORE AND AFTER SMOKE CURE
· RESCUE PARTY ENTERING MINE AFTER EXPLOSION

Entered as second-class matter March 10, 1913, at the postoffice at New York, N. Y., under the act of March 3, 1879.
Copyright, 1913, by The Mentor Association, Inc.



ere it possible for the lump of coal that we burn in our stove, grate or furnace to tell its story, it would take us back millions of years to a time when vast areas of the earth's surface were covered with swamps, supporting a luxuriant vegetation. No human being, mammal or bird yet existed. Animal life included fish, shellfish and other aquatic species, besides reptiles and insects. The vegetal forms resembled our modern ferns, horsetails, club-mosses and evergreens. The atmosphere was heavily charged with moisture, and a mild climate prevailed even in the polar regions. Such were the conditions under which, during the great Carboniferous Age, most of the existing coal-beds were deposited in the earth.

Coal is the litter of primeval swamps and forests. Year after year the débris of the humid jungles accumulated in shallow water or in the boggy soil, where it underwent partial decay, and was thus converted, first of all, into the slimy or spongy material known as "peat." Similar deposits are in process of formation in the swamps of the present day, and the peat obtained from them is dried and used as fuel on an extensive scale in some parts of the world; especially Ireland, Holland, Germany and Scandinavia.

Gradual changes in the elevation of the land led to the submergence of the prehistoric peat bogs, during successive intervals of time, by lakes or shallow seas. Thus their vegetation was killed, and they were overspread with layers of mud or sand, above which, during a subsequent period of elevation, a new peat bog would form; and this process was repeated several times. The conversion of the peat into coal appears to have resulted from the pressure of the overlying strata, probably aided by the internal heat of the earth. Much of its moisture was squeezed



FOREST SWAMP OF THE
CARBONIFEROUS PERIOD (Coal Age). From
a drawing by Potonié and Gothan

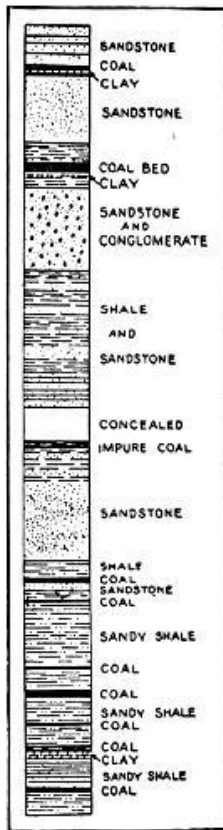


CONCRETE PORTAL OF A "DRIFT" MINE

and evaporated out; the proportions of its component gases were reduced; and the result was a hard mineral, which has earned the popular name of "black diamond" because consisting chiefly of carbon—the same chemical element which, in a pure and crystalline form, constitutes the true diamond. Chemically, coal consists of carbon; the gases hydrogen, nitrogen and oxygen; sulphur; and ash (the mineral matter that remains after combustion).

The record of these long-ago events is found when we sink a shaft through typical coal-bearing strata. We pass through not one, but several, layers of coal, which may vary in thickness from a fraction of an inch to a hundred feet or more, and are separated by generally much thicker layers of sandstone or shale (solidified clay). The layers of coal are known as "coal-beds." Unless a coal-bed is at least two feet thick it is hardly worth working, and, ordinarily, the thickness of a bed does not exceed eight or ten feet. The shale or sandstone above a bed is very commonly found to contain the remains or the impressions of the ancient plants from which the coal was formed. A study of these remains and casts has made it possible to classify hundreds of species of plants now extinct. Fragments of plants are also sometimes found in the coal itself, and thin slices of coal frequently show a vegetable structure under the microscope. Finally, to furnish conclusive proof of the vegetable origin of coal, we find under the coal-bed a layer known as the "underclay," which is a fossil soil filled with the roots and rootlets of the coal-producing plants. Different conditions of formation, and also, probably, differences in the character of the original vegetation, have resulted in the production of different kinds of coal. The most important heat-producing constituent in coal is the elementary substance called "carbon," and the simplest classification of solid fuels depends upon the percentages of fixed (non-volatile) carbon they contain, the average percentages running as follows: Wood, 50%; peat, 55%; lignite, 73%; bituminous coal, 84%; anthracite coal, 93%. When fuel is burned the greater part of it unites chemically with the oxygen of the air to form certain invisible gases—especially carbon dioxide and water-vapor—and only the ash remains.

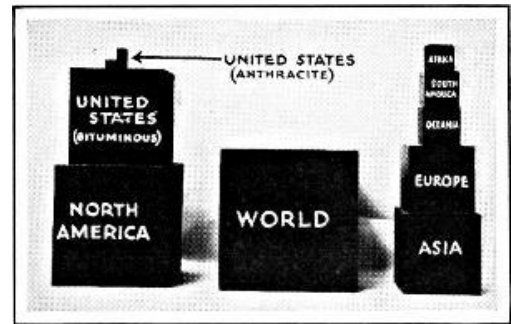
In the popular mind coal is classified as hard or soft, while hard coal is further classified according to the size of the lumps. For both scientific and industrial purposes more elaborate classifications are necessary, and several have been used or proposed.



Courtesy of United States National Museum

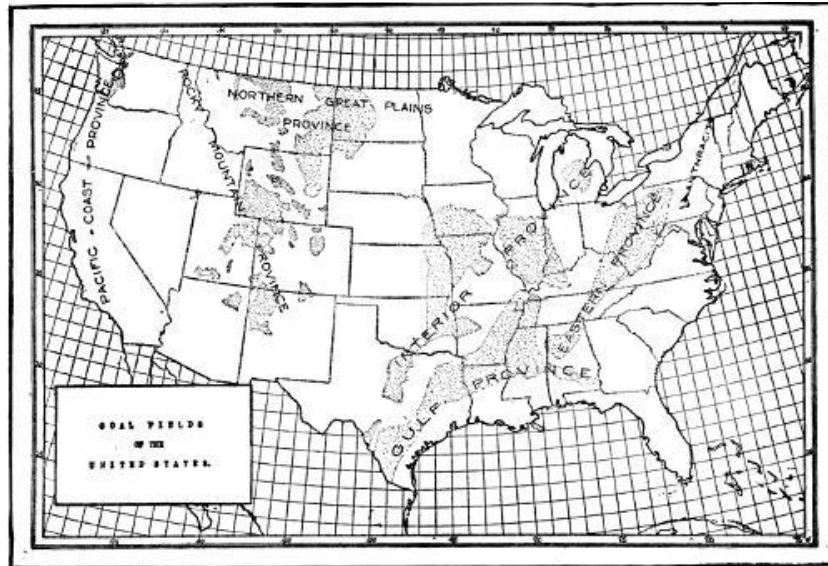
Comparative coal supplies of the world.

The nick in the smallest cube shows how much hard coal has been used up. Soft coal cube has hardly been scratched



From "Geology, Physical and Historical," by H. P. Cleland. American Book Co., N. Y.

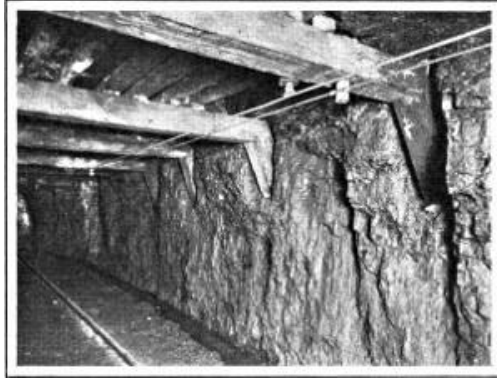
Section of coal-bearing strata in Pennsylvania, showing relative amount of coal and barren rock in a rich field



COAL FIELDS OF THE UNITED STATES

Kinds of Coal

The United States Geological Survey classifies coals, first of all, according to “rank,” depending upon both chemical and physical characteristics. Anthracite, which contains the largest percentage of carbon, ranks highest, and lignite, with the smallest percentage of carbon, lowest. Coals of the same rank are said to be of high or low “grade,” according to whether they contain a relatively small or large percentage, respectively, of ash and sulphur. The ranks recognized by the Survey are: Anthracite, semi-anthracite, semi-bituminous, bituminous, sub-bituminous, and lignite.



Courtesy of U. S. Bureau of Mines

AN "ENTRY" IN A COAL MINE

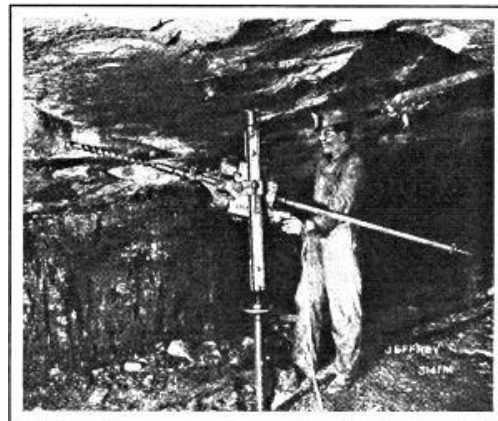
Showing timbered roof



Courtesy of U. S. Bureau of Mines

COATING WALLS OF A MINE WITH CEMENT

To prevent coal-dust explosions



DRILLING IN COAL FOR BLASTING

"Anthracite" is the hardest of coals. It was formed from bituminous coal under the crushing pressure due to the upheaval of mountains or by the intense heat of adjacent molten rocks. Most American anthracite is mined in eastern Pennsylvania. The largest deposits in the world are found in China. Anthracite burns slowly, with little smoke. It is well adapted for domestic use on account of its cleanness, but is not an economical fuel for steam-raising or general manufacturing.

"Semi-anthracite" also ranks as a hard coal, though it is less hard than anthracite. Very little is mined in this country, and it is generally sold as anthracite.

"Semi-bituminous" coal is a softer coal, which, when properly burned, gives off but little smoke. The best semi-bituminous coal ranks highest among the coals in heating value. It is the most valuable fuel for manufacturing purposes; also for steamships, as it requires less bunker space per unit of heat than any other coal.

"Bituminous" coal, or ordinary "soft coal," burns readily, with a smoky flame, and is the coal most commonly used for manufacturing purposes; in fact, the bulk of the coal mined throughout the world belongs to this rank. It includes a good many varieties, some of which

are extensively used in making coke, while others, such as “cannel” coal, have been in great demand for use in gas-works. Nowadays, however, the widespread introduction of “water-gas,”^[2] which does not require any particular kind of coal, has diminished the demand for “gas coals.”

^[2] Made by forcing steam over glowing coal or coke. See Monograph No. 4.

“Sub-bituminous” coal, or “black lignite,” is common in some of our western coal fields. It is a clean and useful domestic fuel when used near the mines, but is not very satisfactory for shipment, as it shrinks and crumbles under the effects of “weathering” and is liable to spontaneous combustion.

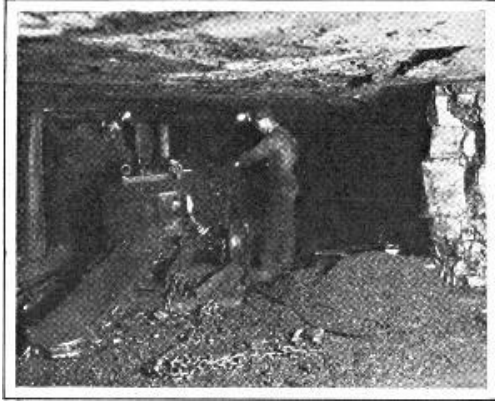
“Lignite” is the least valuable of coals, and is the form of coal which is the least altered from the original peat. The Geological Survey applies this name only to those coals which are distinctly brown and either markedly woody or claylike in appearance. Lignite, as it comes from the mine, contains from thirty to forty per cent. of moisture, and it “slacks” or falls to pieces much more rapidly than sub-bituminous coal when exposed to the air. It is hardly suitable for transportation.

For commercial purposes coal is also classified according to size. The coal as it comes out of the mine, without any sorting into sizes, is known as “run of mine,” and the semi-bituminous coals are commonly shipped in this form. Most coals, however, are passed over bars or gratings, which constitute screens of different degrees of fineness; each screen permits all the lumps below a certain size to fall through, and thus the coal is divided into the different standard sizes. The sizes of anthracite, from the smallest to the largest, are: rice, buckwheat, pea, chestnut (or nut), stove, egg, broken (or grate), steamboat, and lump. Bituminous coal is divided into slack, nut and lump (the largest size). A mixture of lump and nut is called three-quarter coal.



Press Illustrating Service
MODERN ELECTRIC
LOCOMOTIVE
Used for hauling coal from the
mines

The Modern History of Coal



Press Illustrating Service

MODERN MINING MACHINE

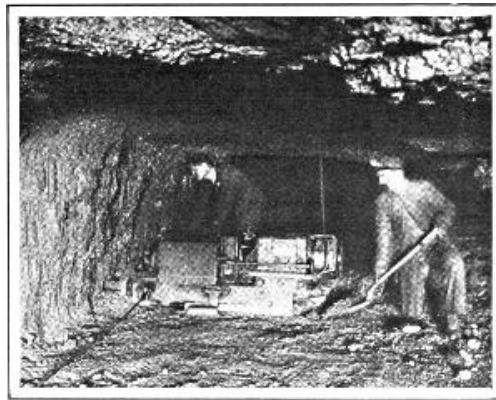
for undercutting coal. The “cutter bar” is shown in front, filled with “cutting teeth” set in a chain that travels around. See illustration opposite



Press Illustrating Service

SHAKER SCREENS IN A TIPPLE HOUSE

The coal passing over screens is graded according to size



“CUTTER BAR” AT WORK

The bar, with its cutting chain of teeth, makes a horizontal cut deep into the bottom of the seam of coal. Blasting then does the rest

The age of the steam-engine is also the age in which the use of coal has become widespread, and the output of coal is a faithful index of industrial progress. Although the Greek writer Theophrastus (about 300 B. C.) mentions the use of coal as a fuel, and its use was also known to the ancient Britons and the Chinese, it was virtually unknown throughout the Middle Ages. The first record of coal mining in England is of the year 1180 A. D., and coal was first shipped to London in the year 1240. It was long known as stone-coal, pit-coal, etc., to distinguish it from charcoal; also as sea-coal, on account of being carried to London by sea. Bituminous coal was first mined in America in 1750, near Richmond, Virginia. Anthracite was discovered in Rhode Island in 1760, and in Pennsylvania in 1766, but for many years its value was not recognized. As late as the year 1812 Colonel George Shoemaker, of Pottsville, was treated as an impostor and threatened with arrest for attempting to sell a few wagon-loads of anthracite in Philadelphia; methods

of burning it were not understood, and it was declared to be merely “black stone.” In the year 1820 only 365 tons of anthracite were sold in this country, as compared with the present annual output of about 90,000,000 tons.

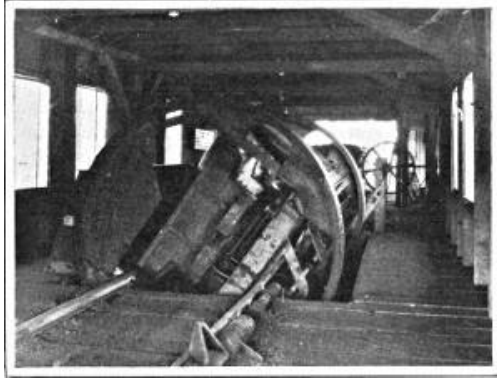
Before the days of the railway coal was shipped mostly by water in rough boats called “arks,” which floated down the rivers to the seaboard towns. As it was impossible to return against the current, the ark was sold with the coal at its destination. A great many arks were wrecked in transit, and the whole process of transportation was a costly one. Only with the introduction of steamboats, canals and railways, did the coal industry assume serious proportions.

The production of coal in America has grown at an amazing rate. In the year 1868 Great Britain produced 3.6 times as much coal as the United States, and the output was also exceeded by that of Germany. In 1899, the United States took the lead. At the present time, with an estimated production for the year 1917 of 643,600,000 tons, the United States is producing nearly half of all the coal mined in the world. Great Britain ranks second, closely followed by Germany.

How Coal is Mined

A relatively small amount of coal is quarried near the surface of the ground from open pits. The overlying soil is removed by steam-shovels, and the coal is then blasted out and shoveled into cars.

Most coal is mined underground. Access to the coal-beds is obtained either by sinking a vertical “shaft” or by driving a tunnel, according to the location of the beds. A tunnel driven at a steep angle is called a “slope.” A horizontal tunnel leading into a coal-seam is called a “drift.” In this country few coal mines are more than 300 or 400 feet below the surface, and the deepest is about 1,600 feet. Much deeper mines are found in Europe, especially in Belgium.



Press Illustrating Service

ROTARY DUMP IN A TIPPLE

Showing a coal car half turned over in order to dump contents



Courtesy of "Coal Age"

SPIRALIZING MACHINES

which, by rotating motion, separate the coal from slate



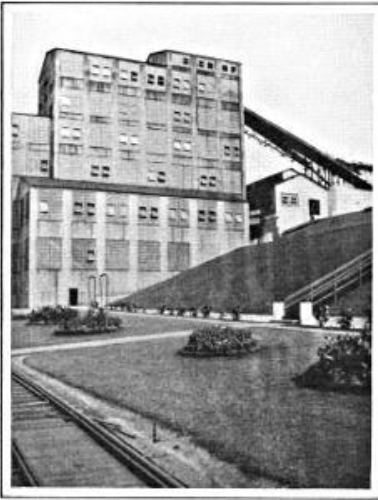
Courtesy of U. S. Bureau of Mines

MODERN HEADFRAME, BINS AND TRESTLE

Of fireproof construction. Anthracite coal mine

American mine shafts are generally rectangular and are divided into two or more compartments. Where a shaft passes through water-bearing strata it must be provided with a tight lining, or "tubbing," to prevent the mine from being flooded. All water that enters the mine collects in an excavation, or "sump," at the bottom of the shaft, and must be pumped to the surface.

The method of working coal-seams most commonly practiced in this country is known as the "room-and-pillar" system. One or more tunnels, or "entries," are first driven from the bottom of the shaft or the mouth of the drift. These are the main thoroughfares of the mine, and are usually provided with tracks, over which the mine cars are hauled by mules or by some other method of traction—locomotives, endless chains, etc. Secondary entries ("headings," "butt entries," etc.) branch off from the main entries. Finally, the work of extracting the coal consists of excavating open spaces, or "rooms," adjoining the entries.



Courtesy of "Coal Age"

MODEL COAL BREAKER

Note the neat and careful "upkeep"
of the place

The actual mining is done in the rooms, and different methods are in use. Anthracite is generally "shot from the solid"; that is, blasted out from the face of the coal without any preliminary cutting. This method is objectionable, especially in bituminous mines (where it is, however, much practiced), because the large charges of powder it requires produce a great deal of coal-dust and weaken the roof and pillars, often leading to falls of coal and fatal accidents. A better plan consists of "undercutting" the coal before it is blasted out. A long groove is made at the level of the floor, either with a pick or with a coal-cutting machine. Holes are then drilled some distance above the groove for the insertion of the blasting charges, and the coal is blasted down. A single shot will sometimes dislodge a ton or two of coal.

The next step is to shovel the coal from the floor into a mine car, which is then pushed into the adjacent entry. The miner attaches a numbered tag to the car, so that he will be duly credited for his work, which is paid for by the ton. The loaded cars are eventually hoisted or hauled out of the mine, to be weighed and discharged above ground.

The final step in working a coal-seam by the room-and-pillar method is to mine out the thick walls or pillars of coal, which are originally left between adjacent rooms to support the roof. As this work proceeds the worked-out sections are filled with waste rock, or the roof is allowed to fall. The object is to leave as little coal in the mine as possible, but practically it is rare that more than 60 or 70 per cent. is recovered.

One feature of a coal mine that must be carefully planned is the system of ventilation. This is provided not merely for the comfort of the miners, but to prevent, as far as possible, the accumulation of poisonous and explosive gases. There are always at least two airways leading into the mine (one or both of which may also be used for hoisting or other purposes), known as the "upcast" and the "downcast," according to the direction in which the air passes through them. A current of air is maintained either by keeping a fire burning at the bottom of the upcast or by the use of powerful fans or blowers. A system of tight trap-doors prevents the air from taking a short cut between the downcast and the upcast, and thus leaving the greater part of the mine unventilated.



Courtesy of "Coal Age"

MINING FROM THE OUTSIDE

Stripping the surface of a coal-bed with steam shovel, at Pittsburg, Kans.



Courtesy of U. S. Bureau of Mines

WATCHING THE CANARY

for indications of poisonous coal gas. Reserves testing the air of a mine after an explosion



Courtesy of U. S. Bureau of Mines

MEMBERS OF RESCUE TEAM

Showing apparatus worn on entering mines after explosions. This device sustains a man for two hours

The coal in the mine constantly gives off various gases, one of which, the notorious "fire-damp" (methane or marsh-gas), is responsible for many explosions. In recent years it has been discovered that coal-dust itself, when mixed with the right proportion of air, is violently explosive. Mine explosions may be minimized by requiring the use of "safety-lamps" (oil, gasoline, or electric); by providing devices to prevent sparking in electrical apparatus; and by using for blasting operations only so-called "permissible" explosives, which give a shorter and cooler flame than black powder. Coal-dust explosions can be largely prevented by wetting the walls of the mine, or by the new process of "rock-dusting," which consists of applying dry incombustible powdered rock to all surfaces. Unfortunately, none of these precautions are employed as generally as they should be.

The elevator used for hoisting in the mine shaft is called a "cage." After the mine cars reach the surface they pass upon an elevated structure called the "tipple." This is generally the most conspicuous feature of a mining property above ground, and provides facilities for screening and otherwise "preparing" the coal as it passes down chutes to the railway cars underneath. The more elaborate structure used for anthracite is called a "breaker"; it includes machinery for crushing the coal and arrangements for removing "slate" and other waste rock by hand picking or otherwise.

Coal mining in this country gives employment to an army of 765,000 men. The word "army" has a sinister appropriateness in this connection, since out of every thousand men employed in the industry three are killed and one hundred and eighty injured annually.

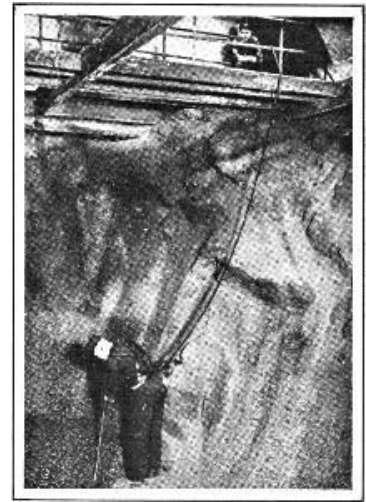
The World's Coal Resources

In order of output, the leading coal-producing countries of the world are: United States, Great Britain, Germany, Austria-Hungary, France, Russia, Belgium, Japan, China, India, and Canada. The total production during the latest year for which data are available was about 1,346,000,000 tons.

How long will the world's coal supply last? This is a question to which various answers have been given. Geologists are able to furnish a rough estimate of the amount of coal now in the ground and near enough to the surface to be mined; but with the growth of the world's industries the demand for coal is increasing by leaps and bounds, and nobody can safely predict how much will be needed at any future time.

The world's "coal reserves"—that is, the amount of coal remaining unmined—are estimated at 8,154,322,500,000 tons. In the United States it is estimated that we have used only four-tenths of one per cent. of our available coal supply. At the *present* rate of consumption the coal in this country would last about 4,000 years; but if the present rate of *increase* in consumption should be maintained, it would last only 100 years!

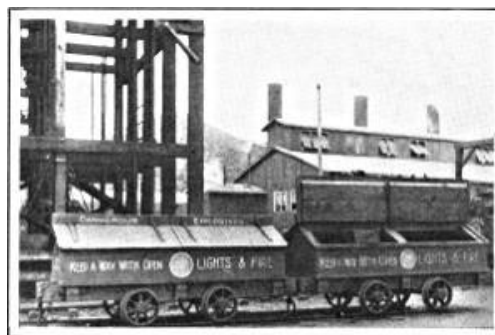
Fortunately for posterity there are sources of heat, light and power which are not, like the fuels, exhaustible. Water-power, for example, is a permanent asset, and there are other inexhaustible sources of energy, such as solar heat and the internal heat of the earth, which Man's ingenuity will someday turn to good account.



Press Illustrating Service

SAFETY DEVICE IN COAL BUNKER

In case of a "coal slide," a man may be pulled out before he is buried and stifled



Courtesy of "Coal Age"

CARS FOR CARRYING EXPLOSIVES INTO MINES

SUPPLEMENTARY READING

COAL CATECHISM

By W. J. Nicholls

THE STORY OF AMERICAN COALS

By W. J. Nicholls

A YEAR IN A COAL MINE

By Joseph Husband

YEAR BOOK OF THE U. S. BUREAU OF MINES

STORY OF A PIECE OF COAL

By E. A. Martin

THE COAL FIELDS OF THE UNITED STATES
(U. S. Geological Survey, Professional Paper 100)

By M. R. Campbell

** Information concerning these books may be had on application to the Editor of The
Mentor.

THE OPEN LETTER

Coal is “a burning question,” that has to be met and answered every day. It supplies heat, light and power—and a thousand and one useful by-products—and it is an ever-present, ever-fruitful subject of public and private discussion. We average folk know something of the varied uses of coal in the big affairs of the world, but we know it more intimately and vitally in the forms in which it ministers to our own personal welfare. Coal, in our everyday—and night—life, means heat and light. It means home comfort—and if this “coal comfort” is denied us, or even curtailed, we raise an immediate and mighty outcry. And why not? The health of a community can be fatally affected by a few heatless days. The experience of the past winter has shown us how dependent we are on fuel, not only for luxury and comfort, but for life itself.



Why do we need so much heat? Many of the peoples of the earth get along comfortably with much less heat than we consider necessary. Europeans and South Americans call us a “steam-heated nation.” Why do we have to surrender so completely and abjectly to the domination of Old King Coal? It is true, as Owen Meredith said: “Civilized man cannot live without cooks”; and light is all important in turning night hours to advantage; but why must we be so warm? Humanity was not created in a warm room, nor was the human race nurtured, in its infancy, by a coal fire or a gas stove. Primitive man was his own heater. He had to *discover* fire, and then exploit its uses. He was originally supplied by nature with a warm body, and he now finds artificial ways of making it warmer. Has not civilization pampered us to a point that has impaired our original heat-giving resources and substituted a forced warmth that has enervated us? The doctors tell us that many diseases come out of artificial heat—indoor diseases, they might be called—the diseases that are treated, and sometimes cured today, by foregoing artificial heat and going back to nature.



Does this mean that I suggest reverting to primitive conditions and giving up heat? No, indeed. I suffered enough last winter. I do not advocate giving up heat—suddenly. But letting up gradually on artificial heat, I do most earnestly advocate. Most of us live an over-heated existence—to the depletion of our health. The steam pipe, like a huge python, is closing its coils about us, and gradually stifling our native vital resources.



On the coldest days of winter a white-haired man, nearly seventy years of age, may be seen walking New York streets, without a hat, clad only in light “Palm Beach” trousers, and

a silk negligée shirt, open at the throat. “He is crazy,” you say. “Perhaps,” I answer, “but at any rate he is healthy—and immune from cold.” Heatless days mean nothing to him. On a raw, drizzling day in November last a slender man was playing golf in a light woolen suit. A companion player, weighing over 200 pounds, full blooded and hearty in appearance, and bundled up in two heavy sweaters, asked the lightly clad player if he was not afraid of catching a fatal cold. “No,” he answered, “*you* are the one that gives me concern. If I had your clothing on I would be a sick man. *I am not healthy enough* to wear all those things.”

★ ★ ★

Which means that we would be better off in health if we could accustom ourselves to less heat; if we could live as the people of some other nations do—comfortable and content with heat enough to take the chill off the air, and not demanding that we shall be “kept going” by means of artificial heat outside of our own natural heat-giving apparatus. We make caloric cripples of ourselves by giving crutches to nature in the form of roaring furnaces and hissing steam pipes. Fresh cold air is better for us than hot air—in winter as well as in summer. Would it not be worth while to form a national Fresh Air Fraternity, based on the principle of foregoing artificial heat and developing the original body caloric? We would then leave artificial heat largely to infants, weaklings and invalids; we would abolish several diseases altogether, improve the mortality rate, and be healthy, happy and vigorous. Incidentally, too, we would have more coal for cooking and other really necessary purposes.


EDITOR

W. D. Moffat
EDITOR

IN THE HIGH SCHOOL

The Mentor is a part of our High School. Each day 192 pupils and 11 teachers have opportunity to partake of its splendid contents. Think of a big family of 200 enjoying it daily. I say daily because I have all the back numbers, and each day a new Mentor with all its gravures is displayed on a specially constructed rack in my recitation room. The important features to be found in it are enumerated on the blackboard, and after having pursued this method for a year I am anxious to tell you something about it.

One sixth of the members of the graduating class at the end of the year chose Mentor subjects for their theses, and, from the supplementary reading list given, selected books, which were obtained by mail from the Carnegie Library at Pittsburgh, with a result that all the theses were most excellent.

The Mentor is not filled with current material, which changes constantly and is good only for a short time. All you print will bear being assimilated and preserved. It is your small statements that make The Mentor big.

It isn't the "new" but the old things it tells, that are not generally known, that gives The Mentor permanent value. It creates a permanent interest in Music, Art, Literature, History, etc., in a High School away from the polish of a big city in such a way as nothing else but a Mentor could. I can't see how a High School can afford to be without a complete set. Make it known to the schools. It will do a great work.

J. B. SHEETZ, Principal, McClellandtown, Penn.

THE MENTOR ASSOCIATION

ESTABLISHED FOR THE DEVELOPMENT OF A POPULAR
INTEREST IN ART, LITERATURE, SCIENCE, HISTORY,
NATURE, AND TRAVEL

THE MENTOR IS PUBLISHED TWICE A MONTH

BY THE MENTOR ASSOCIATION, INC., AT 114-116 EAST 16TH STREET, NEW YORK, N. Y. SUBSCRIPTION, FOUR DOLLARS A YEAR. FOREIGN POSTAGE 75 CENTS EXTRA. CANADIAN POSTAGE 50 CENTS EXTRA. SINGLE COPIES TWENTY CENTS. PRESIDENT, THOMAS H. BECK; VICE-PRESIDENT, WALTER P. TEN EYCK; SECRETARY, W. D. MOFFAT; TREASURER, J. S. CAMPBELL; ASSISTANT TREASURER AND ASSISTANT SECRETARY, H. A. CROWE.

THE MENTOR ASSOCIATION, Inc., 114-116 East 16th Street, New York, N. Y.

Statement of the ownership, management, circulation, etc., required by the Act of Congress of August 24, 1912, of The Mentor, published semi-monthly at New York, N. Y., for April 1, 1918. State of New York, County of New York. Before me, a Notary Public, in and for the State and county aforesaid, personally appeared Thomas H. Beck, who, having been duly sworn according to law, deposes and says that he is the Publisher of The Mentor, and that the following is, to the best of his knowledge and belief, a true statement of the ownership, management, etc., of the aforesaid publication for the date shown in the above caption, required by the Act of August 24, 1912, embodied in section 443, Postal Laws and Regulations, to wit: (1) That the names and addresses of the publisher, editor, managing editor, and business manager are: Publisher, Thomas H. Beck, 52 East 19th Street, New York; Editor, W. D. Moffat, 114-116 East 16th Street, New York; Managing Editor, W. D. Moffat, 114-116 East 16th St., New York; Business Manager, Thomas H. Beck, 52 East 19th Street, New York. (2) That the owners are: American Lithographic Company, 52 East 19th Street, New York; C. Eddy, L. Ettlinger, J. P. Knapp, C. K. Mills, 52 East 19th Street, New York; M. C. Herczog, 28 West 10th Street, New York; William T. Harris, Villa Nova, Pa.; Mrs. M. E. Heppenheimer, 51 East 58th Street, New York; Emilie Schumacher, Executrix for Luise E. Schumacher and Walter L. Schumacher, Mount Vernon, N. Y.; Samuel Untermyer, 120 Broadway, New York. (3) That the known bondholders, mortgagees, and other security holders, owning or holding 1 per cent. or more of total amount of bonds, mortgages, or other securities, are: None. (4) That the two paragraphs next above, giving the names of the owners, stockholders, and security holders, if any, contain not only the list of stockholders and security holders as they appear upon the books of the Company, but also, in cases where the stockholder or security holder appears upon the books of the Company as trustee or in any other fiduciary relation, the name of the person or corporation for whom such trustee is acting, is given; also that the said two paragraphs contain statements embracing affiant's full knowledge and belief as to the circumstances and conditions under which stockholders and security holders who do not appear upon the books of the Company as trustees, hold stock and securities in a capacity other than that of a bona fide owner; and this affiant has no reason to believe that any other person, association, or corporation has any interest direct or indirect in the said stock, bonds, or other securities than as so stated by him. Thomas H. Beck, Publisher. Sworn to and subscribed before me this 18th day of March, 1918. J. S. Campbell, Notary Public, Queens County. Certificate filed in New York County. My commission expires March 30, 1919.

THE MENTOR ASSOCIATION, Inc., 114-116 East 16th St., New York, N.Y.

THE MENTOR

Every Reader of The Mentor

will be interested in knowing that during the past twelve months more than 400,000 back copies of The Mentor were sold to the members of The Mentor Association, which shows, more than anything we might say, the enduring value of every issue of the periodical that is published.

In no other place is it possible to get such a vast amount of information in such condensed, interesting form—and the wonderful gravures that accompany each number are found in no other work. The previous numbers of The Mentor when bound make an invaluable library of knowledge.

We have just received from the binders a number of sets, finished in a beautiful three-quarter morocco. It is complete in five handy volumes, containing all the material that has been published up to and including issue 120. It contains 2,000 text pages, 720 full page gravures, and over 1,800 other rare illustrations. The sets that have just come in are especially well-bound; so strong, in fact, that they will last for years. The leather is an extra fine skin, the clear cut lettering and the cover decorations are in gold—making a set that will prove a handsome addition to any library.

As a member of The Mentor Association you appreciate the value of possessing a work that has the merit of the material that makes up The Mentor Library. Each subject is complete in every detail—and you have the privilege of possessing all subsequent volumes as fast as they are issued, which in time will be the most complete, the most authoritative, and the most interesting set ever published. It is a series that you will be proud to pass on to your children and to your children's children; and the price is so low, and the terms so easy, that you surely must take advantage of it.

YOU NEED SEND NO MONEY NOW. Simply send us your name and address with the request to forward them to you. We will send the five attractive volumes, all charges paid. You can then send us only \$1.00 within ten days after receipt of bill and \$3.00 a month until \$36.25 has been paid—or five per cent. discount for cash, should you choose.

If you will send us a post card or a letter at once we will extend your subscription for one year without additional cost, to begin at the expiration of your present membership; also you will receive, without additional charge (regular price, 25c.), the new cross-reference index, which makes every subject in the library at your instant command.

You Save \$4.25 If You Act Promptly

Several hundred sets of The Mentor Library have been sold during the past two or three weeks, and as we have but a few sets of the present edition on hand, we are making this very unusual offer. It is necessary that you write us promptly that you may take advantage of the present low prices. We urge you to act today.

THE MENTOR ASSOCIATION
114-116 East Sixteenth Street, New York City

MAKE THE SPARE MOMENT COUNT



THE MENTOR

Every Reader of The Mentor

will be interested in knowing that during the past twelve months more than 400,000 back copies of The Mentor were sold to the members of The Mentor Association, which shows, more than anything we might say, the enduring value of every issue of the periodical that is published.

In no other place is it possible to get such a vast amount of information in such condensed, interesting form—and the wonderful gravures that accompany each number are found in no other work. The previous numbers of The Mentor when bound make an invaluable library of knowledge.

We have just received from the binders a number of sets, finished in a beautiful three-quarter morocco. It is complete in five handy volumes, containing all the material that has been published up to and including issue 120. It contains 2,000 text pages, 720 full page gravures, and over 1,800 other rare illustrations. The sets that have just come in are especially well-bound; so strong, in fact, that they will last for years. The leather is an extra fine skin, the clear cut lettering and the cover decorations are in gold—making a set that will prove a handsome addition to any library.

As a member of The Mentor Association you appreciate the value of possessing a work that has the merit of the material that makes up The Mentor Library. Each subject is complete in every detail—and you have the privilege of possessing all subsequent volumes as fast as they are issued, which in time will be the most complete, the most authoritative, and the most interesting set ever published. It is a series that you will be proud to pass on to your children and to your children's children; and the price is so low, and the terms so easy, that you surely must take advantage of it.

YOU NEED SEND NO MONEY NOW. Simply send us your name and address with the request to forward them to you. We will send the five attractive volumes, all charges paid. You can then send us only \$1.00 within ten days after receipt of bill and \$3.00 a month until \$6.25 has been paid—or five per cent. discount for cash, should you choose.

If you will send us a post card or a letter at once we will extend your subscription for one year without additional cost, to begin at the expiration of your present membership; also you will receive, without additional charge (regular price, 25c.), the new cross-reference index, which makes every subject in the library at your instant command.

You Save \$4.25 If You Act Promptly

Several hundred sets of The Mentor Library have been sold during the past two or three weeks, and as we have but a few sets of the present edition on hand, we are making this very unusual offer. It is necessary that you write us promptly that you may take advantage of the present low prices. We urge you to act today.

THE MENTOR ASSOCIATION
114-116 East Sixteenth Street, New York City

**MAKE THE SPARE
MOMENT COUNT**

*** END OF THE PROJECT GUTENBERG EBOOK THE MENTOR:
THE STORY OF COAL, VOL. 6, NUM. 6, SERIAL NO. 154, MAY 1,
1918 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.